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Light occlusion at forest edges: an analysis of tree architectural characteristics

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Abstract

We examined the phenomenon of tree canopy closure by lateral branch expansion at the edges of temperate forest fragments bordering fields, in southwestern Ontario, Canada. We evaluated the relationship between tree crown characteristics and the degree of light reduction (a measure of canopy closure) among six tree species of differing shade tolerance (*Fagus grandifolia* Ehrh., *Acer saccharum* Marsh., *Quercus rubra* L., *Prunus serotina* Ehrh., *Juglans nigra* L. and *Populus grandidentata* Michx.). Tree species differed in their capacity to occlude light: the most effective were the most shade tolerant species, *F. grandifolia* and *A. saccharum*, while the most shade intolerant trees, *J. nigra* and *P. grandidentata*, had the highest percentage of light transmission. The most shade tolerant species presented most of their branches and foliage towards the field and resulted in the most voluminous crowns with the largest values in branch density, branch clumping, canopy depth and canopy width. The best model to predict light occlusion was obtained when we considered the effect of each species as a factor, and foliage biomass facing the field as covariates; this explained 80% of the total variation in light transmission. The ability of shade tolerant trees to enlarge their crowns in response to an increase in light availability allows them to occlude light better. To maximize the protection of microclimates in forest fragments, shade tolerant trees should be encouraged at forest edge communities by favoring recruitment or by selective thinning of the least shade tolerant species.

Keywords: Fragment edges; Lateral branch expansion; Light penetration; Shade tolerant species; Tree canopy characteristics

1. Introduction

The great majority of forest ecosystems in the world have been fragmented by anthropogenic activities converting continuous forests into landscapes of remnant forest patches, embedded in agricultural matrices. The switch from a continuous cover to small fragments brings with it many changes in physical, chemical and abiotic flows in the altered landscape (e.g. Yates et al., 1994; UNEP, 1995; Malcolm, 1998) as well as a suite of new habitat conditions that may be rare or non-existent in continuous forest communities.

The most obvious effects of forest fragmentation are a decreased forest area with increased isolation of the biota and the appearance of permanent, rather than ephemeral, edge habitats. An edge can be considered a transitional ecosystem between the forest and the surrounding open areas, whose creation may induce a changed forest microclimate (Kapos, 1989; Williams-Linera, 1990; MacDougall and Kellman,

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1992; Malcom, 1994; Didham and Lawton, 1999), higher tree mortality (Lovejoy et al., 1986; Kapos et al., 1997), greater tree density (Palik and Murphy, 1990; Williams-Linera, 1990), increased plant growth and foreign invasions (Brothers and Spingarn, 1992; Murcia, 1995) and lower recruitment of herbs (Benitez-Malvido, 1998; Jules, 1998). Avian diversity may also be reduced at the edges by an increased nest predation (Gibbs, 1991; Burkey, 1993; Soderstrom, 1999).

Microclimatic variables, such as air temperature, vapor pressure deficit, and light intensity can vary greatly along an edge-to-interior gradient (Murcia, 1995; Kapos et al., 1997; Didham and Lawton, 1999), due to the increased amount of solar radiation and wind that penetrates into the forest interior (Ranney et al., 1981; Matlack, 1993). Recently created forest fragments possess an open edge where trees lack any significant lateral crown spread and a deep penetration of atmospheric conditions exists (Matlack, 1993). However, trees appear to respond to increased light levels at these boundaries and eventually, edges with trees exhibiting extensive lateral crowns develop (Fig. 1). At long-established edges, these well-developed lateral canopies appear to promote effective closure and may insulate the forest interiors from edge effects (Brothers and Spingarn, 1992; MacDougall and Kellman, 1992). As a result, atmospheric edge effects may become considerably attenuated with time.

Tree species may vary in their capacity to expand their branches laterally. Sakai (1987) and Canham et al. (1994) suggested that those species that have larger crowns, in terms of radius, may have better lateral branching growth in the northern temperate zone. Such trees include species of *Acer*, *Carya* and *Fagus* which are classed as shade tolerant. In contrast, less shade tolerant species appear to be less prone to developing and sustaining lateral canopies.

Although many mathematical models that simulate light interception and crown characteristics have been developed (e.g. Kurth, 1994; Niklas, 1994; Hilbert and Messier, 1996; Pearcy and Yang, 1996; Takenaka et al., 1998), the adaptive and functional significances of lateral branching at permanent forest edges have received far less systematic discussion (for exceptions, see Canham et al. (1994) and Millet et al. (1998)).

The aim of the present study was to examine tree canopy closure at the edges of forest fragments and to evaluate whether there are significant differences between species that could affect their capacity to insulate the fragment from atmospheric edge effects. We ask how species that evolved in forests without permanent edges respond to this new habitat. We present two hypotheses: (1) Species of greater shade

Fig. 1. Lateral tree canopy development by sugar maple (*A. saccharum*) and beech (*F. grandifolia*) at a long-established forest edge in southern Ontario. The foliage wall is formed by lateral branches from individual trees developed throughout their full height.

tolerance will close an edge more effectively than those of less shade tolerance. (2) Tree species will vary in their capacity to close an edge as a consequence of differences in branching patterns, foliage density and foliage arrangement. To test these hypotheses, the degree of light reduction was used as a measure of canopy closure, and we analyze the relationship between tree architectural characteristics and light occlusion among tree species of differing shade tolerance.

2. Methods

2.1. Site description and tree selection

The study was conducted in southwestern Ontario, along the northern shore of lake Erie (42°39′ N, 80°33′ W), on long-isolated fragments immersed in an agricultural landscape. The "Carolinian Forest", a species-rich temperate deciduous forest, comprises the natural vegetation in this area.

Mature edges, comprised of trees exhibiting lateral branching throughout their trunk, were surveyed in May 1998 to identify a population of trees for architectural analysis. Species were sought which included a variety of shade tolerances and were sufficiently aggregated to enable all focal trees to be surrounded by conspecifics. For practical reasons, the study focused on trees of intermediate size (15–24 m tall) rather than the largest individuals encountered at edges. Six species (*Fagus grandifolia*, *Acer saccharum*, *Quercus rubra*, *Prunus serotina*, *Juglans nigra* and *Populus grandidentata*) were selected; these species met the above criteria and provided a sample size of 10 individuals per species.

Most forest fragments in the area are rectangular in shape, following the original land survey, and possess edges facing in the four cardinal directions. Because edge orientation has an effect on microclimatic conditions, all trees selected were located on straight forest edges of South, East and West aspects (Table 1). Individuals facing North were avoided so that all edge canopies could be assumed to have a comparable exposure to solar radiation (Matlack, 1993; Murcia, 1995). Sixty adult trees (10 per species) were sampled during the summer period May–August 1998.

2.2. Light measurement

Light transmitted through individual canopies was measured using the Ozalid paper technique (Friend, 1961; Sullivan and Mix, 1982). Ozalid paper sensors were calibrated in photosynthetically active photon flux density (PPFD) using a LICOR 190SA quantum sensor (Licor, Lincoln, Nebraska, USA) connected to a LI-COR 1000 data logger; this yielded a close linear relationship between these two measures ($r^2 = 0.97$, $N = 25, P < 0.001$).

Stacks of photosensitive Ozalid paper were placed in petri dishes, which were attached horizontally to small wooden platforms screwed into the lowest canopy branch that extended approximately perpendicularly from the edge to the field. This location for light measurement ensured that the effects of each focal tree's crown was maximized and that interference from neighboring trees and understory vegetation was minimized. In a pilot study, a series of

Table 1
Summary of general characteristics of the tree population sampled[a]

	A. saccharum (sugar maple)	*F. grandifolia* (American beech)	*Q. rubra* (red oak)	*P. serotina* (black cherry)	*J. nigra* (black walnut)	*P. grandidentata* (big-tooth aspen)
Shade tolerance[b]	High	High	Intermediate	Low	Low	Very low
Edge aspect	South- and east-facing	East- and west-facing	East- and west-facing	South-facing	East-facing	South- and west-facing
Tree height (m)	22.54 (3.78)	19.81 (2.99)	19.69 (5.82)	18.57 (4.87)	17.79 (2.55)	17.44 (3.31)
DBH (cm)	35.84 (3.43)	38.63 (15.17)	26.64 (11.59)	24.76 (5.11)	33.54 (6.41)	24.28 (3.49)
Age (years)	46.4 (5.79)	77.8 (28.31)	28.4 (11.17)	24.1 (6.82)	33.8 (9.61)	30 (8.58)

[a] For numerical variables, means and standard errors (in parentheses) are given.
[b] Shade tolerance classes according to Burns and Honkala (1990).

instantaneous light measurements made along several branches from the tree stem towards the crown edge detected a steep drop in light levels within 50 cm of the main trunk, presumably as a result of the shading by the trunk itself. Consequently, two petri dishes were mounted on each branch at 50 and 100 cm from the main stem.

Integrated light values beneath each tree crown were measured over two 24 h periods and averaged. External light levels were measured concurrently using the PPFD sensor and data logger. All beneath-crown measurements were expressed as a percentage of external light.

The values of light were log-transformed (to improve normality) and differences in light at 50 and 100 cm were tested with paired *t*-tests. This showed that in two species (*F. grandifolia* and *P. serotina*), light levels at 50 cm were significantly lower than those at 100 cm, presumably because of shading by the trunk. Consequently, all subsequent analyses used only light measured at 100 cm from the trunk, which was assumed to be responding primarily to the tree's crown.

2.3. Tree architectural characteristics

The architecture of trees can be described at different levels of detail. In this study we have stressed characteristics that are likely to influence light transmission and those that can be measured fairly rapidly in the field. We hypothesized that light levels transmitted through tree canopies would be influenced primarily by crown volume and foliage biomass. Because light is captured primarily by the leaf tissue of tree canopies, the quantity of leaf tissue (i.e. its biomass), the arrangement of the leaves within the crown and the leaf biomass distribution throughout the tree should affect light interception (Pearcy and Yang, 1996).

To estimate these crown attributes, three types of measurements, were taken in the field. We ranked them from the simplest ones that were rapidly measured (gross characteristics), to the more sophisticated (e.g. leaf area index, LAI) and the most time consuming: foliage biomass. Because foliage biomass represents a variable whose measurement in the field is laborious, we also explored the architectural variables that would allow an expeditious but accurate estimation of this characteristic.

2.3.1. Gross architectural characteristics

We predicted that greater crown depths (cdept) and widths would be more effective in reducing light. A large number of branches (nbr) may be expected to support a greater amount of biomass, so a highly branched tree would sustain more foliage biomass than a tree with very few branches. Furthermore, clumping of branches may be expected to increase foliage self-shading and reduce the amount of light transmission compared to regularly spaced branches.

Six simple architectural features were measured in the field: (i) total height and diameter at breast height (dbh) of each tree, (ii) the maximum width of the tree crown, extending into adjacent field, measured along five equally spaced radii at 30°, 60°, 90°, 120° and 150° from the forest edge, (iii) height and orientation of each individual branch of the tree and (iv) diameter of each individual branch at 10 cm from the main trunk.

Tree height, internode and branch lengths were measured with a hand-held laser range-finder (Vertex A5731, CFE Equipment, Edmonton, Canada). Diameter and azimuth of each branch were recorded by a climber. Branch orientation was measured with a field compass and expressed as the angular orientation of each branch with respect to the forest edge. Age of each tree was estimated from cores taken with an increment borer. Because the determination of the age of the oldest trees was uncertain due to heart rot, trees were classified into five age categories of 20 years each.

From the previous measurements we derived a measure of tree size (the ratio between height and dbh, htdbh) and four branching and crown characteristics for each tree, that may influence light transmission:

1. Crown depth (cdept), calculated as the difference between total height and the height of the lowest alive branch.
2. Branch density (brden), as the number of branches per meter of height.
3. An index of branch clumping (clump), as the variance/mean^2 ratio of internode lengths (Ezcurra et al., 1992; Greig-Smith, 1983).
4. Crown width (crwidth), represented by the mean of five width measurements per tree.

2.3.2. LAI and canopy cover

LAI and canopy cover were obtained from hemispherical photography using a fish eye lens (Sigma 8 mm f2.8). LAI is defined herein as the amount of leaf area per unit of ground area and it is a measurement highly correlated to canopy cover. Canopy cover was measured as the total cover of the sky hemisphere in the hemispherical photograph. Except for six trees (two *J. nigra*, two *P. serotina* and two *P. grandidentata*), where the camera was placed at 2 m above ground level, the camera was positioned at the same locations as the petri dishes, leveled with a self-leveling camera mount and aligned to the magnetic North. Photographs were taken on a uniformly overcast day, at the end of July, where the leaves were fully developed. "Hemiphots" were analyzed using Winphot–Hemiphot software (ter Steege, 1994).

Although hemispherical photography has been widely used to estimate light under a forest canopy (e.g. Ackerly and Bazzaz, 1995; Mitchell and Whitmore, 1993; Turner, 1990), this technique was not applied here because of the difficulties in positioning the camera at the base of each tree's canopy, particularly when the lowest branches were higher than 4 m. In contrast, the ozalid technique provided greater flexibility in location of sensors.

2.3.3. Foliage biomass

On the assumption that leaf biomass would provide a better predictor of light transmission than those mentioned above, we estimated foliage biomass in the field. Ten branches covering a range of diameters were cut from each species among the 60 individuals. Foliage was removed, oven-dried to constant mass at 70°C for 3 days and subsequently weighed. The logarithm of leaf weight was regressed against the logarithm of diameter for each branch. Highly significant relationships were obtained for each species: slopes ranged from 1.40 to 3.51 and $0.89 < r^2 < 0.98$. We calculated total leaf biomass for each individual (tbiom), and divided total biomass into three equal depth classes: biomass at the lower (blc), mid (bmc) and upper canopy (buc). We also estimated canopy asymmetry, as the percentage of the total biomass facing outwards at the forest edge (bout).

To adjust leaf biomass for leaf transmissivity, we measured light transmission through 10 leaves per species, at noon, on a clear day, with a LI-COR 190SA quantum sensor, located 2 cm below each leaf or leaflet. Leaf transmissivity was expressed as a percentage of incident light and this value was used to pro-rate foliage biomass for light transmissivity. All subsequent values of foliage biomass includes this adjustment.

2.4. Statistical analysis

2.4.1. Species architectural characteristics

For graphic representation of the orientation data, branches were classified into 12 azimuth classes of 30° each. Branch and foliage frequencies for each individual tree were arranged in a one-way contingency table where branches and foliage were classified into two azimuthal categories of 180° each: facing the field and facing the forest. Differences in branch and foliage orientation between species were tested using log-linear models with the *G*-statistic (McCullagh and Nelder, 1989).

We performed a series of univariate one-way ANOVAs to evaluate differences in canopy depth and crown width between species. Differences in branch density between species were evaluated by means of a Kruskal–Wallis rank test (Hollander and Wolf, 1973).

2.4.2. Light measurements and regression analyses

Untransformed light data confirmed that the assumption of normality was not appropriate. Although different transformations (log, arcsine, reciprocal) of the light variable were tried, the gamma distribution provided the best regression models. We choose generalized linear models with gamma errors and reciprocal link to perform the regression analysis (McCullagh and Nelder, 1989). In the first one, we calculated the relative effect of the different species on light reduction. According to our model, we calculated the relative efficiency in light reduction between species, as a ratio between the estimates (Crawley, 1993). In the second model, the architectural characteristics for the 60 individuals were regressed against light values. We used a multiple regression technique, following a stepwise procedure. In order to decrease Type I error, once a variable had entered the regression model, we did not consider variables that had significant collinearity with it. In the last model, species as a factor and the highest significant predictor from the second regression model, used as a covariate, were regressed against light levels.

The fit of the models was assessed by F-tests and we use a maximum-likelihood method to estimate model parameters. As in standard regression, we tested the residuals of the models for compliance to the error assumptions.

A separate analysis was conducted with foliage biomass as a response variable and simple architectural variables regressed against this. Because variance in biomass increased with the mean and the error was not normally distributed, we also dealt with these data by specifying a gamma error in the regression model. All the analyses were performed in the statistical packages GLIM (NAG, 1985; Crawley, 1993) and SAS (SAS, 1989).

3. Results

3.1. Architectural characteristics

Tree heights ranged from 23.57 to 15.13 m and dbh from 58.8 cm in the largest beech to 13.7 cm in the smallest oak tree. Beech trees stood out as the oldest trees that were sampled (Table 1). Although the 60 chosen individuals were approximately the same height ($F_{5,54} = 1.95$, $P = 0.10$), the population comprised individuals that differed by species in dbh ($F_{5,54} = 6.28$, $P < 0.001$) and age (Table 1).

Foliage orientation varied significantly among species ($G = 12.916$, d.f. $= 5$, $P = 0.024$). In *A. saccharum*, *F. grandifolia*, *Q. rubra* and *P. serotina* most of the foliage faced the field (Fig. 2), while in other species it was more uniformly distributed in all compass directions. *P. grandidentata* and *J. nigra* had nearly the same percentage of foliage facing the field and facing the forest. As with foliage distribution data, a significant preference towards the field in branch distribution was found for *A. saccharum*, *F. grandifolia*, *Q. rubra* and *P. serotina* ($G = 13.78$, d.f. $= 5$, $P < 0.025$).

Significant differences were found in branch density between species (Kruskal–Wallis' $H = 15.425$, d.f. $= 5$, $P = 0.0087$; Fig. 3a). *F. grandifolia*, *P. serotina* and *A. saccharum* had the highest values while *J. nigra* and *P. grandidentata* had the smallest. A similar tendency could be observed for the branch clumping. *A. saccharum*, *F. grandifolia* and *P. serotina* displayed the highest clumping index (Fig. 3b). *Q. rubra*, *J. nigra* and *P. grandidentata* tended to have the lowest branch clumping values.

A one-way ANOVA showed significant inter-specific differences for crown depth and crown width ($F_{5,54} = 3.72$, $P = 0.0058$ and $F_{5,54} = 8.20$, $P < 0.0001$, respectively). For crown depth, *A. saccharum* and *Q. rubra* had the highest means while *P. grandidentata* and *J. nigra*, formed a distinct group with the shortest crown depths (Fig. 3c). *F. grandifolia* and *P. serotina* overlapped with the previous two groups of species. *F. grandifolia* is the species with the longest lateral branches facing the field (Fig. 3d). This species differs significantly from the other five. In contrast, *P. grandidentata*, possessed the shortest lateral branches.

When the means of the variables are ranked, the most shade tolerant species consistently had the largest values in branch density, branch clumping, crown depth and crown width. However, shade intolerant *P. serotina* had branch density and branch clumping, comparable to that of the shade tolerant trees while *Q. rubra*, the only tree of intermediate shade tolerance, was like a shade tolerant tree in crown depth and crown width but was intermediate in branch density and branch clumping.

3.2. Transmission of light through tree canopies

Tree species varied significantly in the amount of light transmitted to the base of their crowns ($F_{5,54} = 19.9$, $P < 0.0001$; Fig. 4). *A. saccharum* occluded the largest proportion of sunlight through its crown, allowing only 3.73% of external light to be transmitted. In contrast, *J. nigra* exhibited the least light occlusion (Table 2). As shown in Table 2 and according to the parameter estimates from the gamma regression model, the efficiency in blocking light could be ranked as follows: *A. saccharum* > *F. grandifolia* > *P. serotina* > *Q. rubra* > *P. grandidentata* > *J. nigra*. In terms of its effect on reducing light, an *F. grandifolia* or *A. saccharum* tree was approximately equivalent to 3 *P. grandidentata* plants or to 5–6 *J. nigra* trees. *A. saccharum* and *F. grandifolia* stood out as the most effective light occluders, followed closely by *P. serotina* and *Q. rubra*. *P. grandidentata* and *J. nigra*, were the least effective in occluding light.

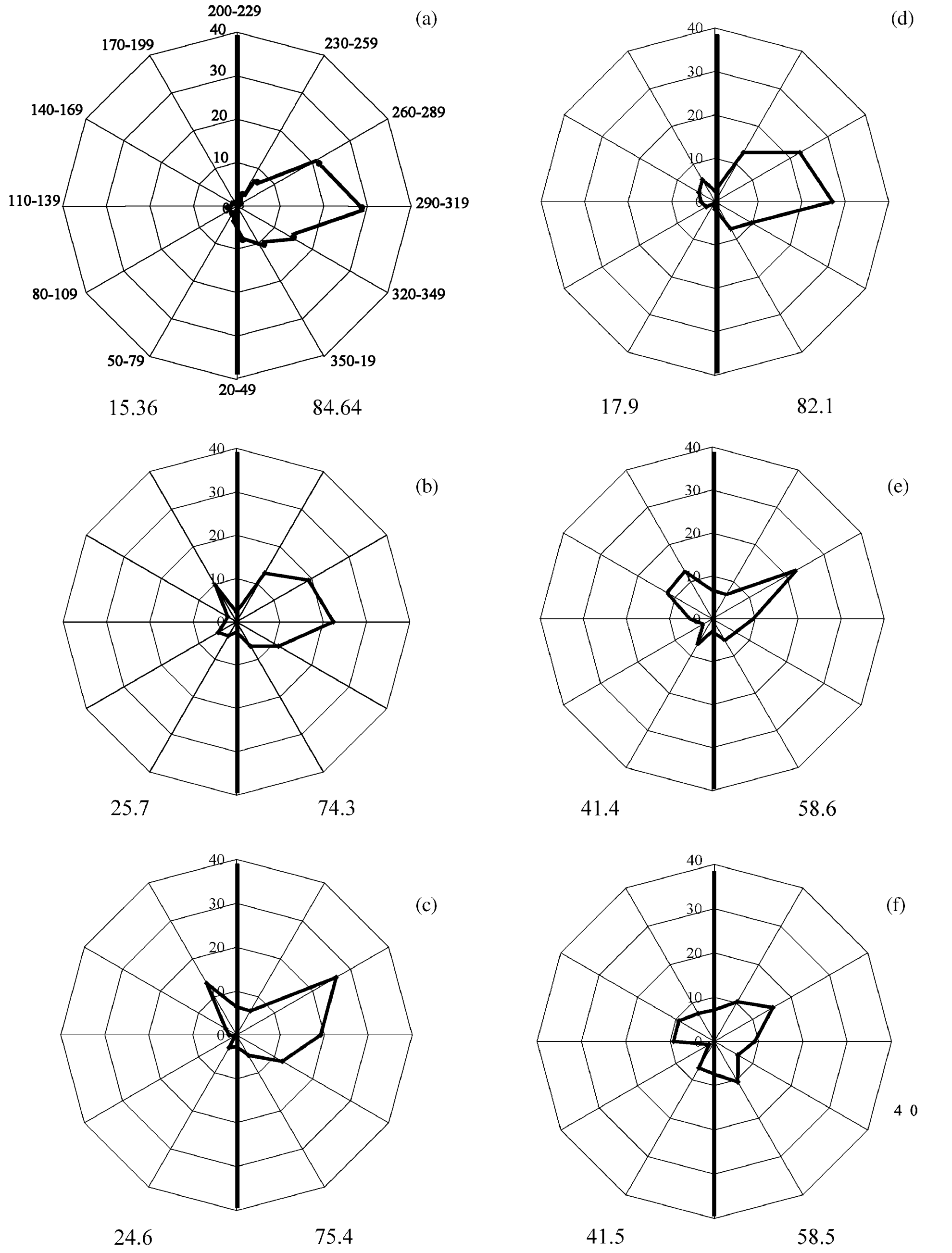

Fig. 2. Radial distribution of foliage density for the six species: (a) *F. grandifolia*; (b) *A. saccharum*; (c) *Q. rubra*; (d) *P. serotina*; (e) *J. nigra*; (f) *P. grandidentata*. Forest is to the left and field to the right of the vertical axes. The radial axes represent the mean percentage of foliage density that grows along each direction for the 10 individuals. Numbers at the right bottom of each chart indicates the percentage of foliage facing the field and the numbers at the left, the percentage of foliage facing the forest interior.

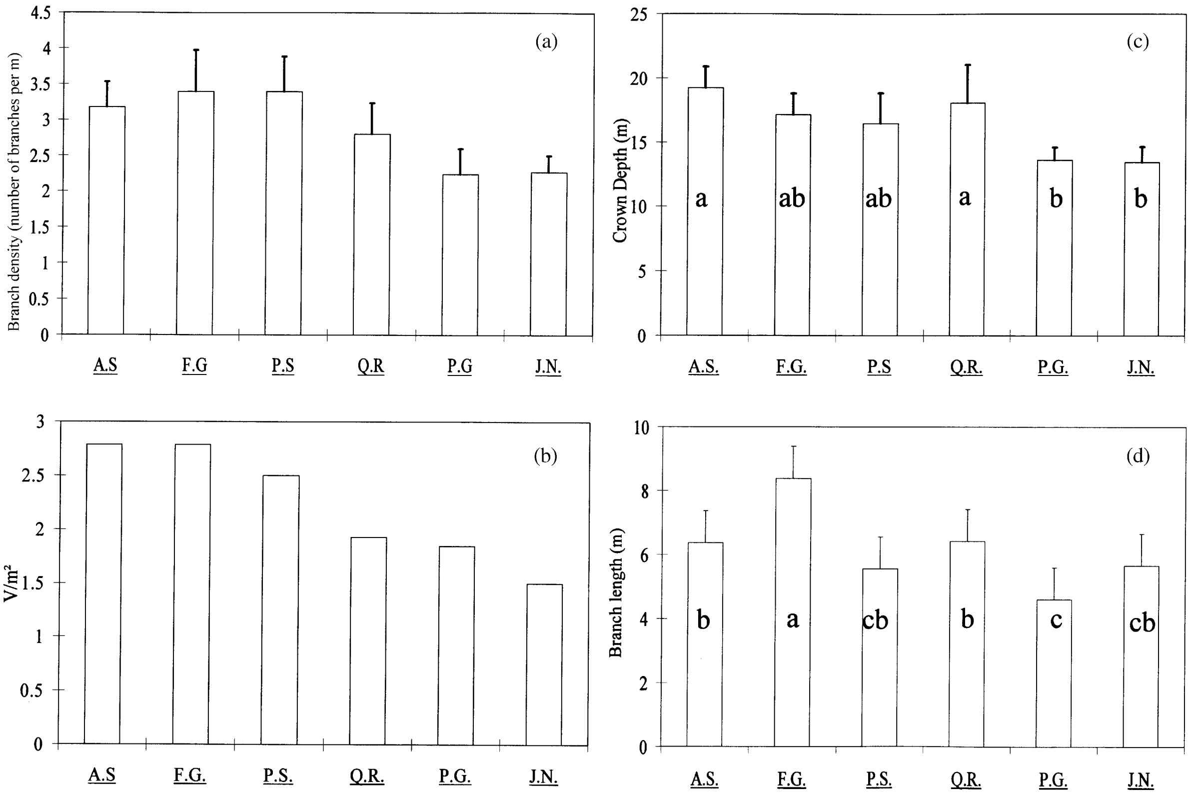

Fig. 3. Architectural characteristics for the six species: *FG* corresponds to *F. grandifolia*, *PG* to *P. grandidentata*, *PS* to *P. serotina*, *AS* to *A. saccharum*, *QR* to *Q. rubra* and *JN* to *J. nigra*. (a) Branch density (±S.E.M.); (b) branch clumping index (or coefficient of variation); (c) canopy depth (±S.E.M.); (d) canopy width (±S.E.M.). In (c) and (d), species with the same letters are not significantly different at $P < 0.05$.

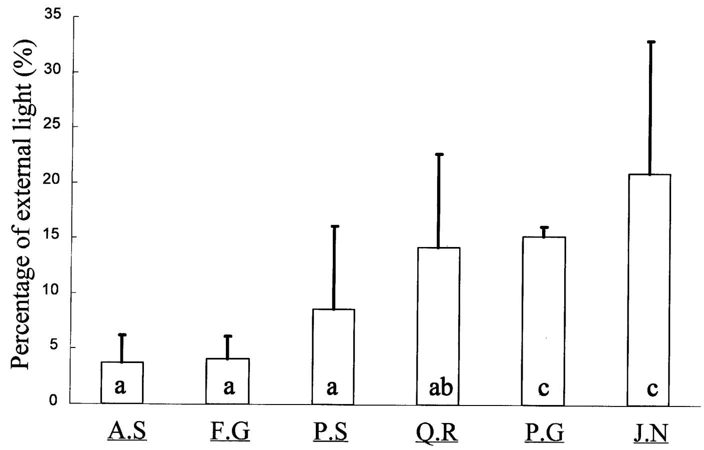

Fig. 4. Percentage of external light transmitted by the tree canopies of the six species (±S.E.M.): *FG* corresponds to *F. grandifolia*, *PG* to *P. grandidentata*, *PS* to *P. serotina*, *AS* to *A. saccharum*, *QR* to *Q. rubra* and *JN* to *J. nigra*. Species with the same letter are not significantly different at $P < 0.05$.

Table 2
Summary of species architectural and light transmission characteristics[a]

	A. saccharum (sugar maple)	*F. grandifolia* (American beech)	*Q. rubra* (red oak)	*P. serotina* (black cherry)	*J. nigra* (black walnut)	*P. grandidentata* (big-tooth aspen)
Crown width (m)	6.37 (1.01)	8.39 (2.28)	6.43 (1.88)	5.56 (0.70)	5.60 (1.12)	4.60 (1.14)
Crown depth (m)	19.21 (3.32)	17.13 (3.34)	18.08 (5.95)	16.47 (4.75)	13.45 (2.36)	13.6 (1.94)
Branch density (per meter)	3.39 (0.57)	3.17 (0.35)	2.80 (0.43)	3.38 (0.49)	2.27 (0.22)	2.23 (0.36)
Branch clumping	2.79	2.79	1.92	2.50	1.50	1.85
LAI[b]	2.073 (0.5)	1.26 (0.15)	1.34 (0.24)	1.73 (0.12)	1.26 (0.21)	1.52 (0.25)
Bout[c] (kg)	27.14 (15.52)	26.07 (15.59)	5.67 (2.14)	7.79 (2.03)	4.97 (1.21)	5.02 (1.01)
Tbiom[d] (kg)	35.33 (14.09)	36.00 (26.02)	15.63 (12.57)	12.44 (3.83)	15.49 (6.67)	13.16 (5.25)
% Light transmitted	3.73 (2.44)	4.11 (1.99)	14.20 (8.46)	8.59 (7.50)	20.95 (12.04)	15.23 (0.88)

[a] Values shown are means and standard errors (in parentheses).
[b] Leaf area index.
[c] Biomass facing the field.
[d] Total biomass.

3.3. Effect of crown architecture on light transmission

When we considered the 60 trees collectively, the best predictor of light reduction beneath tree crowns at forest edges was the amount of foliage biomass which faced the edge (Table 3). This variable explained 66% of the total variance in light transmission. Although an increase in number of branches and a high LAI reduced the amount of light transmitted, the two variables inter-related to biomass, were marginally significant and added only a minor proportion to the total variance explained (Table 3). Neither tree age, height nor dbh were significant predictors for light occlusion.

Setting aside the foliage biomass variables, a model which used crude architectural variables, revealed that the number of branches as the best predictor of light reduction, followed by crown width and LAI (Table 4). The three variables together explained less total variance (45%) than the previous model, which included the foliage biomass predictors.

Similar results were obtained when we modeled the best predictors for external foliage biomass (Table 5). Excluding species as a factor, the number of branches, crown width and LAI explained all together 66% of the variance in biomass, with number of branches being the strongest predictor among the three variables. Including species in our model, this factor accounted for 77% ($F_{5,54} = 35.62$, $P < 0.00001$) of the total variation in biomass, suggesting that foliage density could be architecturally constrained by the species.

However, the best model for predicting light occlusion was obtained when we considered each species as

Table 3
Architectural variables tested as predictors of light reduction

Dependent variable	Best predictors	Percent of the variance explained	*F*	*P*
Light	Bout[a]	66	122.25	<0.00001
	Nbr[b]	2	3.46	0.08
	LAI[c]	2	3.37	0.07
Total		70		

[a] Biomass facing the field.
[b] Number of branches.
[c] Leaf area index.

Table 4
Crude architectural variables tested as predictors of light, excluding biomass variables

Dependent variable	Best predictors	Percent of the variance explained	*F*	*P*
Light	Nbr[a]	24	24.42	<0.00001
	LAI[b]	11	11.65	0.0012
	Crwidth[c]	10	10.58	0.0019
Total		45		

[a] Number of branches.
[b] Leaf area index.
[c] Crown width.

Table 5
Best predictors for external biomass

Dependent variable	Best predictors	Percent of the variance explained	*F*	*P*
External biomass	Nbr[a]	43	68.03	0.00001
	Crwidth[b]	13	20.64	0.0003
	LAI[c]	10	15.07	0.00028
Total		66		

[a] Number of branches.
[b] Crown width.
[c] Leaf area index.

Table 6
Proportion of the variation accounted by the best predictors of the best regression model, including species as a factor

Dependent variable	Best predictors	Percent of the variance explained	*F*	*P*
Light	Sp[a]	61	17.2	<0.00001
	Bout[b]	13	9.7257	0.00294
	Sp × bout	5	4.93	0.00101
Total		79		

[a] Species.
[b] Biomass facing the field.

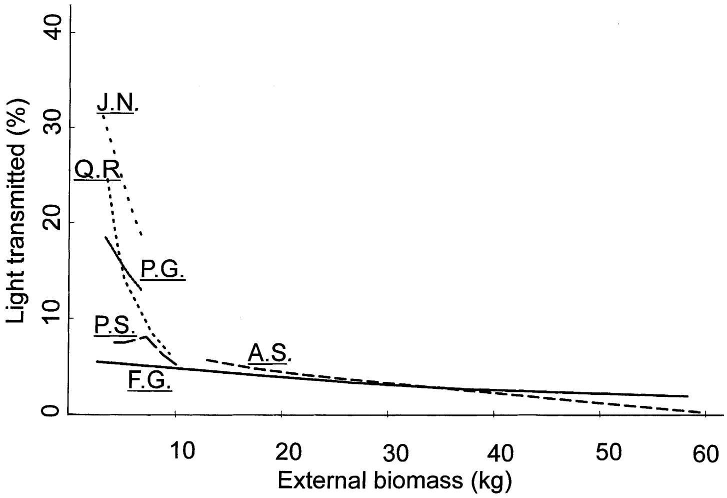

Fig. 5. Relationship between light and external foliage biomass for each species, according to the gamma fit. A smoother LOWESS was used to draw the regression functions (Cleveland, 1979). *FG* corresponds to *F. grandifolia*, *PG* to *P. grandidentata*, *PS* to *P. serotina*, *AS* to *A. saccharum*, *QR* to *Q. rubra* and *JN* to *J. nigra*.

a factor, and biomass facing the field as a covariate (Table 6). To test the significance of the interaction term between these two variables, we asked if the impact of biomass on light was the same for different species. The only significant interactions could be observed for *P. serotina* and *Q. rubra*, suggesting that *P. serotina* and *Q. rubra* have an additional efficiency in blocking light, as foliage biomass increases. When light transmitted is regressed on external foliage biomass, *Q. rubra* and *J. nigra* were found to differ from the other four species: for small changes in foliage biomass, there was an important decrease in the amount of light transmitted (Fig. 5). For *A. saccharum* and *F. grandifolia*, the most efficient crown closers, light transmitted through the crown remains almost constantly low within a wide range values of biomass. *J. nigra* and *P. grandidentata* could not reduce light to the low levels achieved by *A. saccharum* and *F. grandifolia*.

4. Discussion

4.1. Relationship between light transmission, crown architectural features and shade tolerance levels

The six tree species differed in their capacity to occlude light. The two most effective light occluders were also the most shade tolerant species, *A. saccharum* and *F. grandifolia*. In contrast, *J. nigra* and *P. grandidentata*, the most shade intolerant trees, had the highest percentages of light transmission. *Q. rubra*, a tree species of intermediate shade tolerance, demonstrated intermediate levels of light penetration through the crown. As a consequence, a strong relationship seems to exist between the degree of light occlusion and the level of shade tolerance. An apparent exception to this rule is *P. serotina*, a shade intolerant tree which was an efficient light blocker. Furthermore, the effectiveness of occluding sunlight penetration at forest edges by *P. serotina* was greater than that of *Q. rubra*, an intermediate light occluder. Most *P. serotina* branches were oriented outwards which was a characteristic of the shade tolerant species. However, the branches are overhanging, suggesting that unlike the most shade tolerant trees, *P. serotina* was not capable of sustaining long lateral branches. Although *P. serotina* trees were among the youngest sampled, *Q. rubra* for a comparable size and age, was capable of maintaining long lateral branches.

Foliage biomass was closely linked to a reduction in light levels. As an intuitive rule, better developed crowns should occlude more light. Highly branchy trees with wide canopies and a large LAI will result in voluminous trees that will exhibit an efficient canopy closure at forest edges. Any more detailed analysis of the effects of the crown shape on light occlusion would need to take into account the effects that foliage distribution, and branch and leaf inclination, have on this factor at the edges of forest fragments. In our study, the most shade tolerant species exhibited the largest crown size and volume, and the greatest values in foliage biomass facing the field, while the two least shade tolerant species, *J. nigra* and *P. grandidentata*, presented the smallest canopy volume. Are patterns of canopy growth therefore related to the levels of shade tolerance? Based on allometric relationships, crown width, crown depth and branch density, could be considered variables highly inter-related within an individual ontogeny. As a tree ages, we expect an increase in the values of all these variables, resulting in an increase of foliage biomass. However, shade tolerant species and shade intolerant ones differ in their growth pattern and in the capability to allocate resources during shoot growth: Sakai (1987) reported that *Acer* seedlings of high shade tolerance concentrated more on producing an efficient photosynthetic area than on height elongation. On the contrary, shade intolerant *Acer* saplings tended to exhibit a greater vertical growth. Recently, Chen et al. (1996) recognized a similar shoot growth pattern in coniferous saplings of differing shade tolerance. Under reduced light levels, the more shade tolerant *Pseudotsuga menziesii* (Mirb.), allocated more of its production to lateral expansion rather than stem growth, while the same flexibility was not found in the less shade tolerant *Pinus contorta* Dougl. ex Loud. These findings indicate that a growth pattern plasticity is more likely to be present in shade tolerant saplings, and possibly, in the adults. Shade tolerant species seem capable of switching from a vertical to a horizontal growth habit, in response to changes in light and space competitive environment. It is also well known (Givnish, 1984) that shade intolerant species are pre-adapted to rapid vertical elongation until they obviously reach the biophysical limits on height growth: later, success might be determined by their ability to spread horizontally. However, any growth strategy implies construction and maintenance costs and photosynthetic gains (benefits). For example, there is some evidence that the cost to support a horizontal crown spread could be higher than that of a vertical one (Kuuluvainen, 1992). If this is the case, how do shade tolerant species compensate for this cost at forest edges?

Although we have omitted in our architectural analyses one of the most important foliage arrangement characteristics, leaf inclination, we have obtained highly significant models for light occlusion and for foliage biomass. While measurement of foliage biomass is a laborious procedure, our results also show that the other more easily measured variables explain much of the variation in foliage biomass and light occlusion at forest edges: crown width, number of branches and LAI. These variables could provide a means of rapidly comparing the light-occlusion capacities of tree species in regions whose tree biotas are poorly known.

4.2. The development of protective forest edges: implications for management and conservation of forest fragments

The penetration of external atmospheric conditions into forest remnants is frequently seen as one of the most severe threats to the integrity of forest communities in fragmented landscapes (Saunders et al., 1991; Williams-Linera et al., 1998). While PAR is but one of several atmospheric conditions that may be altered by the appearance of a permanent forest edge, Matlack (1993) has demonstrated that most other edge-related environmental changes are closely associated with this, making PAR a useful indicator of aggregate environmental change in closed-canopy forest systems. The results of this study indicate that the development of lateral canopies by trees at long-established edges can play a major role in creating a protected forest interior. Although light was measured on the outer sides of edge trees, and so was influenced by only a single crown layer, it was reduced by some species to less than 5% of external PAR, which is within the range of values that have been recorded in the interiors of temperate deciduous forests (Matlack, 1993).

Deep intrusions of external atmospheric conditions have been documented at recently created forest edges, but long-established edges often demonstrate far less severe intrusions. For example, Kapos (1989) found increased light levels extending to 40 m into recently created forest fragments in Amazonia, but MacDougall and Kellman (1992) found that at long-established tropical gallery forest edges, the zone of elevated light levels had shrunk to only 10–12 m. Comparably low light levels are reported near long-established forest edges in temperate deciduous forests by Brothers and Spingarn (1992) and Matlack (1993). However, the trajectory from the exposed conditions at recent forest edges, to the much more protected conditions at long-established edges, remains poorly documented, as does the long-term dynamics of the forest edge tree community.

Ranney et al. (1981) suggested that canopy closure over a 70-year period in southeastern Wisconsin involved a combination of lateral tree crown expansion and understory development, with tree crowns eventually assuming the dominant role. However, they acknowledged that even the oldest edges observed were unlikely to have achieved an equilibrium. Edge changes over a 12-year period at a tropical forest site in Panama involved primarily release of shade tolerant advanced regeneration to form an understory wall 5–8 m in height, while mature trees showed lateral branch dieback on the exposed edges, rather than crown expansion (Williams-Linera, 1990). At long-established tropical gallery forest edges a complex wall of canopy vegetation usually develops, comprised of a variety of species and growth forms (Kellman et al., 1994). This contrasts with the simplicity of temperate forest edges, where the crowns of single species usually dominate the full depth of the foliage wall (Fig. 1). In the present study, limited evidence of epicormic branching was observed at recent edges, and most lateral branching on edge trees appeared to have developed during tree maturation, as the largest and longest branches were usually those lowest on the tree stem.

The results of this study suggest that the most protective forest edge community in the local area would consist of one dominated by shade tolerant trees of the regional flora. However, it is unclear whether such a community can be expected to develop and persist spontaneously, or whether management intervention would be necessary to promote this. The oldest permanent forest edges in the study area tended to be dominated by shade tolerant trees, and more light demanding species were observed only at expanding or recently disturbed edges. This suggests that the local forest edges may be undergoing normal successional change towards increased dominance by shade tolerant species, comparable to that which prevails in the forest generally. A study of trees in old-growth deciduous forest in Indiana by Brothers (1993) found no major compositional differences between edge and interior zones, although an increased density and species diversity of saplings in the forest understory near edges. This suggest that, with time, edges may become generally favorable zones for the recruitment and growth of a broad spectrum of tree species, rather than becoming increasingly marginal habitats for tree growth. Similar conclusions have been reached about tropical forest edge zones (Kellman et al., 1998). This understory complexity near forest edges implies a potential for complex micro-successions to occur at the gaps created when forest edge trees die, especially if these involve better-illuminated sites than those found in forest interior gaps. Any manipulation to favor recruitment of shade tolerant trees at forest edges would therefore probably most profitably be directed toward selective thinning of advance regeneration in canopy edge zones to release the stems of more shade tolerant species.

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